

NT-C-003 | Data Science | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & AI
Target learners	Graduates and professionals seeking entry-level data-science or machine-learning work.
Prerequisites	Basic mathematics and computer use; Python foundation may be required.
Approved duration	Approved working range: 20-24 weeks / 220-280 guided hours.
Final outcome	Prepare data, build and evaluate machine-learning models, and communicate evidence-based findings through a reproducible project.
Assessment profile	Data/AI

Approved tools / platforms

Python, Jupyter, NumPy, pandas, SQL, matplotlib/seaborn, scikit-learn, Git; deployment tool to be confirmed.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-003 | Data Science | NT-C-003-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Python, NumPy, notebooks; pandas, files, SQL, missing values
Completion evidence	Data manipulation exercises; Clean analysis dataset
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Python for data

Objective: Python, NumPy, notebooks

Observable evidence: Data manipulation exercises

Concept	Approved knowledge statement	Application behaviour	Evidence
Python	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when python is appropriate; apply it in a Data Science task; verify the result.	Data manipulation exercises
NumPy	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when numpy is appropriate; apply it in a Data Science task; verify the result.	Data manipulation exercises
notebooks	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when notebooks is appropriate; apply it in a Data Science task; verify the result.	Data manipulation exercises

Module 2 - Data handling

Objective: pandas, files, SQL, missing values

Observable evidence: Clean analysis dataset

Concept	Approved knowledge statement	Application behaviour	Evidence
pandas	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pandas is appropriate; apply it in a Data Science task; verify the result.	Clean analysis dataset
files	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when files is appropriate; apply it in a Data Science task; verify the result.	Clean analysis dataset
SQL	A language for defining, querying and controlling relational data. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when sql is appropriate; apply it in a Data Science task; verify the result.	Clean analysis dataset
missing values	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when missing values is appropriate; apply it in a Data Science task; verify the result.	Clean analysis dataset

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Data manipulation exercises; Clean analysis dataset, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-003 | Data Science | NT-C-003-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Descriptive statistics, probability, sampling; Patterns, outliers, charts, storytelling
Completion evidence	Statistical analysis; EDA report
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Statistics

Objective: Descriptive statistics, probability, sampling

Observable evidence: Statistical analysis

Concept	Approved knowledge statement	Application behaviour	Evidence
Descriptive statistics	Methods for describing variation and making evidence-based inferences. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when descriptive statistics is appropriate; apply it in a Data Science task; verify the result.	Statistical analysis
probability	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when probability is appropriate; apply it in a Data Science task; verify the result.	Statistical analysis
sampling	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when sampling is appropriate; apply it in a Data Science task; verify the result.	Statistical analysis

Module 4 - EDA & visualization

Objective: Patterns, outliers, charts, storytelling

Observable evidence: EDA report

Concept	Approved knowledge statement	Application behaviour	Evidence
Patterns	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when patterns is appropriate; apply it in a Data Science task; verify the result.	EDA report
outliers	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when outliers is appropriate; apply it in a Data Science task; verify the result.	EDA report
charts	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when charts is appropriate; apply it in a Data Science task; verify the result.	EDA report
storytelling	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when storytelling is appropriate; apply it in a Data Science task; verify the result.	EDA report

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Statistical analysis; EDA report, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-003 | Data Science | NT-C-003-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Regression, classification, preprocessing; Feature engineering, tuning, cross-validation
Completion evidence	Baseline predictive model; Compared models
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Supervised learning

Objective: Regression, classification, preprocessing

Observable evidence: Baseline predictive model

Concept	Approved knowledge statement	Application behaviour	Evidence
Regression	Supervised modelling of a continuous outcome. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when regression is appropriate; apply it in a Data Science task; verify the result.	Baseline predictive model
classification	Supervised modelling of a categorical outcome. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when classification is appropriate; apply it in a Data Science task; verify the result.	Baseline predictive model
preprocessing	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when preprocessing is appropriate; apply it in a Data Science task; verify the result.	Baseline predictive model

Module 6 - Model improvement

Objective: Feature engineering, tuning, cross-validation

Observable evidence: Compared models

Concept	Approved knowledge statement	Application behaviour	Evidence
Feature engineering	Creating or transforming model inputs to represent useful information. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when feature engineering is appropriate; apply it in a Data Science task; verify the result.	Compared models
tuning	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tuning is appropriate; apply it in a Data Science task; verify the result.	Compared models
cross-validation	Checking input or output against defined rules before trusting it. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when cross-validation is appropriate; apply it in a Data Science task; verify the result.	Compared models

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Baseline predictive model; Compared models, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-003 | Data Science | NT-C-003-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Clustering, dimensionality reduction; Pipelines, evaluation, ethics, simple deployment
Completion evidence	Segmentation project; Reproducible capstone
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Unsupervised learning

Objective: Clustering, dimensionality reduction

Observable evidence: Segmentation project

Concept	Approved knowledge statement	Application behaviour	Evidence
Clustering	Unsupervised grouping based on similarity. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when clustering is appropriate; apply it in a Data Science task; verify the result.	Segmentation project
dimensionality reduction	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dimensionality reduction is appropriate; apply it in a Data Science task; verify the result.	Segmentation project

Module 8 - Delivery

Objective: Pipelines, evaluation, ethics, simple deployment

Observable evidence: Reproducible capstone

Concept	Approved knowledge statement	Application behaviour	Evidence
Pipelines	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pipelines is appropriate; apply it in a Data Science task; verify the result.	Reproducible capstone
evaluation	User acceptance testing that validates a solution against agreed business needs. In Data Science, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when evaluation is appropriate; apply it in a Data Science task; verify the result.	Reproducible capstone
ethics	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ethics is appropriate; apply it in a Data Science task; verify the result.	Reproducible capstone
simple deployment	A defined competency within Data Science. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when simple deployment is appropriate; apply it in a Data Science task; verify the result.	Reproducible capstone

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Segmentation project; Reproducible capstone, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-003 | Data Science | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-003-SK01	Python for data	Prerequisite foundation	Data manipulation exercises	Critical
NT-C-003-SK02	Data handling	M01 plus earlier foundations	Clean analysis dataset	Critical
NT-C-003-SK03	Statistics	M02 plus earlier foundations	Statistical analysis	Supporting
NT-C-003-SK04	EDA & visualization	M03 plus earlier foundations	EDA report	Supporting
NT-C-003-SK05	Supervised learning	M04 plus earlier foundations	Baseline predictive model	Supporting
NT-C-003-SK06	Model improvement	M05 plus earlier foundations	Compared models	Critical
NT-C-003-SK07	Unsupervised learning	M06 plus earlier foundations	Segmentation project	Supporting
NT-C-003-SK08	Delivery	M07 plus earlier foundations	Reproducible capstone	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-003 | Data Science | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-003-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Python for data and Data handling?	The learner should provide Data manipulation exercises; Clean analysis dataset and explain how the work applies Python, NumPy, notebooks; pandas, files, SQL, missing values. Recall alone is insufficient.
NT-C-003-S01-EX02	Scenario/remediation	A learner reports completing Data handling but cannot produce Clean analysis dataset. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-003-S01, request Clean analysis dataset, and allow a new equivalent test only after remediation.
NT-C-003-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Statistics and EDA & visualization?	The learner should provide Statistical analysis; EDA report and explain how the work applies Descriptive statistics, probability, sampling; Patterns, outliers, charts, storytelling. Recall alone is insufficient.
NT-C-003-S02-EX02	Scenario/remediation	A learner reports completing EDA & visualization but cannot produce EDA report. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-003-S02, request EDA report, and allow a new equivalent test only after remediation.
NT-C-003-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Supervised learning and Model improvement?	The learner should provide Baseline predictive model; Compared models and explain how the work applies Regression, classification, preprocessing; Feature engineering, tuning, cross-validation. Recall alone is insufficient.
NT-C-003-S03-EX02	Scenario/remediation	A learner reports completing Model improvement but cannot produce Compared models. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-003-S03, request Compared models, and allow a new equivalent test only after remediation.
NT-C-003-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Unsupervised learning and Delivery?	The learner should provide Segmentation project; Reproducible capstone and explain how the work applies Clustering, dimensionality reduction; Pipelines, evaluation, ethics, simple deployment. Recall alone is insufficient.
NT-C-003-S04-EX02	Scenario/remediation	A learner reports completing Delivery but cannot produce Reproducible capstone. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-003-S04, request Reproducible capstone, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-003 | Data Science | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Sales or operations EDA	Cleaning, statistics, visualization	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Customer or student segmentation	Clustering and interpretation	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Predictive model with documented business use	Pipeline, metrics, communication	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-003 | Data Science | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior Data Scientist	Python 34% [critical]; statistics 33% [critical]; ML 33%	Evaluated model and report	If a critical skill is below 60 or evaluated model and report is missing, return Improve First or Needs Evidence with the linked module/project.
2	Machine Learning Analyst	Preprocessing 34% [critical]; modelling 33% [critical]; metrics 33%	Reproducible notebook/project	If a critical skill is below 60 or reproducible notebook/project is missing, return Improve First or Needs Evidence with the linked module/project.
3	Data Science Associate	EDA 25% [critical]; SQL 25% [critical]; visualization 25%; ML 25%	Portfolio capstone	If a critical skill is below 60 or portfolio capstone is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-003 | Data Science | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Data Science for?	Graduates and professionals seeking entry-level data-science or machine-learning work.
What should I know before starting?	Basic mathematics and computer use; Python foundation may be required.
How long is the approved learning journey?	Approved working range: 20-24 weeks / 220-280 guided hours.
What will I be able to demonstrate?	Prepare data, build and evaluate machine-learning models, and communicate evidence-based findings through a reproducible project.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Data Science knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-003; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```